Q $y = \text{arcsec } x$ $\quad \frac{dy}{dx}?$

A $\text{arcsec } x = y$

$\sec(y) = x$

$\frac{d}{dx}$ ↓

$\sec(y)\tan(y) \cdot \frac{dy}{dx} = 1$

$\frac{dy}{dx} = \frac{1}{\sec(y)\tan(y)}$

We can replace $\sec(y)$ with x but what about $\tan(y)$?

$\sin^2 y + \cos^2 y = 1$

$\frac{\sin^2 y}{\cos^2 y} + \frac{\cos^2 y}{\cos^2 y} = \frac{1}{\cos^2 y}$

$\tan^2 y + 1 = \sec^2 y$

$\tan^2 y = \sec^2 y - 1$

$\tan^2 y = \pm\sqrt{\sec^2 y - 1}$

$\tan(y) = \sqrt{\sec^2 y - 1}$

$\frac{dy}{dx} = \frac{1}{\sec(y)\sqrt{\sec^2(y) - 1}}$

$\frac{dy}{dx} = \frac{1}{x\sqrt{x^2 - 1}}$